

Data-driven control of linear systems using quantized data

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Abstract

This paper studies data-driven stabilization of unknown discrete-time linear systems using only quantized state measurements that take values in finite sets. The proposed approach consists of two stages. In the controller design stage, a dynamic quantization scheme is employed to collect data with bounded error, and a semidefinite program (SDP) is formulated from these measurements. We show that any feasible solution to the SDP yields a stabilizing feedback gain, provided that the solution satisfies a condition involving the quantization error bound and the collected data. In the feedback stabilization stage, the quantizer is updated via a Lyapunov-based design so that the number of quantization levels becomes fixed and the closed-loop system under quantized state feedback converges exponentially. A key feature of the proposed method is that the number of quantization levels remains finite in both stages and constant during stabilization. Simulation results illustrate the effectiveness of the approach.

I. INTRODUCTION

Data-driven methods provide powerful tools for controlling systems where accurate models are unavailable or difficult to obtain [1], [2]. Indirect approaches based on system identification followed by model-based control design may suffer from model mismatch, which can compromise closed-loop stability and performance guarantees. In contrast, direct data-driven methods bypass explicit model identification and design controllers directly from data [3], [4]. A key development in this direction is Willems' fundamental lemma [5], which shows that the behavior of a linear system can be fully characterized using data from a sufficiently excited trajectory. This result has led to a behavioral framework for data-driven analysis and control of linear systems [6], [7], with applications to linear quadratic regulation [6], [8] and model predictive control [9], and extensions to nonlinear systems [10] and switched systems [11]. In this paper, we adopt this framework for data-driven controller design.

Modern control systems are increasingly implemented over digital communication networks, where data must be transmitted through rate-limited channels [12]. This has led to extensive research on control under data-rate constraints. A common approach is to quantize measurements into finite sets before transmission, thereby enabling finite data rates [13]. Foundational results using this approach are surveyed in [12], [14], with applications in areas such as power electronics, robotics, and aerospace [15]. Recent developments span areas including robust control [16], [17], event-triggered control [18], and switched systems [19], [20].

These considerations motivate the study of data-driven control under data-rate constraints. Most existing data-driven control methods rely on high-resolution measurements, which limits their applicability in communication-constrained environments. Extensions to noisy data have introduced robustness guarantees using tools such as semidefinite programming [8], the matrix S-lemma [21], and data informativity [22]. However, these approaches do not explicitly address data-rate constraints.

Data-driven control under quantized measurements remains relatively underexplored. In [23], stabilization of unknown linear systems with quantized control inputs is studied using logarithmic quantizers. Related works consider superstabilization with quantized inputs and states [24], and event-triggered control under quantization that achieves input-to-state stability (ISS) and its practical variants [25]. The trade-off between quantization density and control performance is analyzed in [26]. However, these methods rely on quantizers with infinitely many levels, resulting in infinite data rates. Moreover, [23]–[25] compute feedback gains using unquantized data, thereby bypassing a central challenge in data-driven control under data-rate constraints.

In contrast, we develop a data-driven framework for stabilizing unknown discrete-time linear systems using only quantized state measurements that take values in finite sets. The proposed approach consists of two stages.

First, in the controller design stage, a dynamic quantizer maps state measurements to finite sets while maintaining a quantization error bound. A semidefinite program (SDP) is formulated from these measurements. We

show that any feasible solution to the SDP yields a stabilizing feedback gain, provided that the solution satisfies a condition involving the quantization error bound and the collected data. The number of quantization levels may vary over time but remains finite and can be determined locally.

Second, in the feedback stabilization stage, the quantizer is updated using a Lyapunov-based design so that the number of quantization levels becomes fixed. Under this design, the closed-loop system with the feedback gain computed in the first stage and quantized state measurements is shown to converge exponentially to the equilibrium.

A key feature of the proposed method is that the number of quantization levels (and hence the required data rate) is finite at all times and constant during stabilization. This establishes a direct connection between data-driven control and data-rate constraints, demonstrating that stabilization of unknown systems can be achieved under communication constraints.

The remainder of the paper is organized as follows. Section II introduces the system model, control objective, and basic quantization scheme. Section III develops the data-driven controller design and establishes conditions for obtaining stabilizing feedback gains. Section IV presents the quantized feedback stabilization and proves exponential convergence. Section V provides simulation results, followed by concluding remarks in Section VI.

Notations: For a vector $v \in \mathbb{R}^n$, denote by $\|v\|_2$ and $\|v\|_\infty$ its Euclidean norm and ∞ -norm, respectively, which satisfy

$$\|v\|_\infty \leq \|v\|_2 \leq \sqrt{n}\|v\|_\infty. \quad (1)$$

For a matrix $M \in \mathbb{R}^{n \times m}$, denote by $\|M\|_2$ and $\|M\|_F$ its spectral norm and Frobenius norm, respectively, which satisfy $\|M\|_2 \leq \|M\|_F$. For a symmetric matrix $P \in \mathbb{R}^{n \times n}$, denote by $\bar{\lambda}(P)$ and $\underline{\lambda}(P)$ its largest and smallest eigenvalues, respectively.

II. PROBLEM FORMULATION

Consider a discrete-time linear time-invariant (LTI) system

$$x(k+1) = Ax(k) + Bu(k), \quad x(0) = x_0, \quad k \in \mathbb{Z}_{\geq 0}, \quad (2)$$

where $x(k) \in \mathbb{R}^n$ is the state and $u(k) \in \mathbb{R}^m$ is the control input. We assume that the dimensions n and m are known, but the matrices $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times m}$ are unknown. Instead, only upper bounds on their norms are known:

$$\|A\|_2 \leq \alpha, \quad \|B\|_2 \leq \beta. \quad (3)$$

Our goal is to stabilize the unknown system (2) using quantized state measurements $\hat{x}(k) = q_k(x(k))$, where $q_k(\cdot)$ is a time-varying quantizer that maps $\mathbb{R}^n$ to a *finite* set of points at each time k .

The proposed approach consists of two stages. First, during the *controller design stage*, we employ a data-driven framework to compute a stabilizing feedback gain $K \in \mathbb{R}^{m \times n}$ from input and quantized state data collected along the current trajectory. The quantizer update rule for this stage is designed to keep the quantization error bounded. Second, during the *feedback stabilization stage*, we introduce an adjusted quantizer update rule and establish exponential convergence of the closed-loop system under quantized state feedback $u(k) = K\hat{x}(k)$.

The basic quantization scheme is introduced in Section II-A. The controller design stage is presented in Section III, and the feedback stabilization stage is presented in Section IV.

A. Quantization scheme

The initial state x_0 is unknown. We assume that there exists a known bound $E_0 > 0$ such that

$$\|x_0\|_\infty \leq E_0.$$

If such a bound is unavailable, it can be obtained in finite time using standard search techniques in quantized control; see, e.g., [13].

At each time k , we maintain a bound E_k such that

$$\|x(k)\|_\infty \leq E_k, \quad (4)$$

that is, $x(k) \in R_k$, where

$$R_k := \{x \in \mathbb{R}^n : \|x\|_\infty \leq E_k\}. \quad (5)$$

The hypercube R_k serves as the quantization range. Given E_k , update rules are designed to compute E_{k+1} such that $x(k+1) \in R_{k+1}$. The update strategies differ in the two stages and are derived in Sections III-B and IV-A.

The quantizer partitions R_k into N_k^n equal hypercubic cells, with N_k cells per dimension, where N_k is a positive integer. The quantized measurement $\hat{x}(k)$ is defined as the center of the cell containing $x(k)$. Consequently, the quantization error

$$d(k) := x(k) - \hat{x}(k)$$

satisfies

$$\|d(k)\|_\infty = \|x(k) - \hat{x}(k)\|_\infty \leq \frac{E_k}{N_k}. \quad (6)$$

Since the quantizer takes values in a set of cardinality N_k^n , the measurement at time k can be transmitted using a finite number of bits.

III. DATA-DRIVEN CONTROLLER DESIGN

A. Data-driven control preliminaries

During the controller design stage, we excite the unknown system (2) with an input sequence $\{u(k)\}_{k=0}^{T-1}$ satisfying

$$\|u(k)\|_2 \leq \bar{u}, \quad \forall k \in \{0, 1, \dots, T-1\}, \quad (7)$$

and collect the quantized state sequence $\{\hat{x}(k)\}_{k=0}^T$, where

$$T \geq m(n+1) + n. \quad (8)$$

Define the data matrices

$$\begin{aligned} \hat{X}_0 &:= [\hat{x}(0) \quad \hat{x}(1) \quad \dots \quad \hat{x}(T-1)], \\ \hat{X}_1 &:= [\hat{x}(1) \quad \hat{x}(2) \quad \dots \quad \hat{x}(T)]. \end{aligned} \quad (9)$$

for the quantized state measurement. The data matrices $U_0 \in \mathbb{R}^{m \times T}$ for the input $u(k)$, $X_0, X_1 \in \mathbb{R}^{n \times T}$ for the unquantized state $x(k)$, and $D_0, D_1 \in \mathbb{R}^{n \times T}$ for the quantization error $d(k)$ are defined similarly; X_0, X_1, D_0, D_1 are introduced only for analysis and are not available for control synthesis.

A key rank condition for computing a stabilizing controller is

$$\text{rank} \left(\begin{bmatrix} U_0 \\ \hat{X}_0 \end{bmatrix} \right) = n + m. \quad (10)$$

In the absence of quantization, it is known from [5] that this rank condition holds when the input sequence is sufficiently exciting.

Definition 1 ([5]). The input sequence $\{u(k)\}_{k=0}^{T-1}$ is *persistently exciting (PE)* of order L if the Hankel matrix

$$\mathcal{U}_{0,L,T-L+1} := \begin{bmatrix} u(0) & u(1) & \dots & u(T-L) \\ u(1) & u(2) & \dots & u(T-L+1) \\ \vdots & \vdots & \ddots & \vdots \\ u(L-1) & u(L) & \dots & u(T-1) \end{bmatrix}$$

has full rank mL .

Lemma 1 ([5, Cor. 2]). If the system (2) is controllable and the input sequence $\{u(k)\}_{k=0}^{T-1}$ is PE of order $n+1$, then

$$\text{rank} \left(\begin{bmatrix} U_0 \\ X_0 \end{bmatrix} \right) = n + m. \quad (11)$$

The lower bound on T in (8) is necessary for $\{u(k)\}_{k=0}^{T-1}$ to be PE of order $n+1$. In Theorem 2 below, we show that the quantized rank condition (10) holds if, additionally, the quantization errors are sufficiently small. The proof is based on a robust extension of Lemma 1 from [27].

B. Quantized data collection

For $k < T$, using the bounds (3), (4), and (7) together with (1), we obtain

$$\begin{aligned}\|x(k+1)\|_\infty &\leq \|Ax(k) + Bu(k)\|_2 \\ &\leq \|A\|_2 \sqrt{n} \|x(k)\|_\infty + \|B\|_2 \|u(k)\|_2 \\ &\leq \alpha \sqrt{n} E_k + \beta \bar{u}.\end{aligned}$$

To ensure that (4) holds for all $k \leq T$, we let

$$E_{k+1} := \alpha \sqrt{n} E_k + \beta \bar{u}, \quad k = 0, 1, \dots, T-1. \quad (12)$$

To ensure that quantization error is bounded by a prescribed constant $\delta > 0$, we let

$$N_k := \left\lceil \frac{E_k}{\delta} \right\rceil, \quad k = 0, 1, \dots, T, \quad (13)$$

where $\lceil \cdot \rceil$ is the ceiling function. Then (6) implies

$$\|d(k)\|_\infty \leq \delta \quad \forall k \in \{0, 1, \dots, T\}. \quad (14)$$

Note that the update rule (12) does not depend on data but leads to exponential growth of E_k and N_k during this stage. To mitigate this conservativeness, one can use

$$E_{k+1} := \alpha \left(\|\hat{x}(k)\|_2 + \frac{\sqrt{n} E_k}{N_k} \right) + \beta \bar{u} \quad (15)$$

for $k = 0, 1, \dots, T-1$. By (6) and the triangle inequality, this update rule also ensures that (4) holds for all $k \leq T$. Moreover, it utilizes the collected data $\hat{x}(k)$ and typically leads to a significant reduction in the required number of quantization cells in practice.

Although the number of quantization cells (and hence the number of bits required for transmission) varies during this stage, it can be computed locally at the sensor, enabling a finite data transmission rate. The number of quantization cells is constant during the feedback stabilization stage in Section IV.

C. Data-driven control synthesis

We adopt the optimization-based framework for stabilizing unknown LTI systems in the presence of disturbances from [8] and extend it to quantized state measurements.

In the absence of quantization, a stabilizing controller can be computed by solving the following *semidefinite program (SDP)*:

$$\begin{aligned} &\min_{(\gamma, Q, P, L)} \gamma \\ &\text{subject to} \\ &\begin{cases} X_1 Q P^{-1} Q^\top X_1^\top - P + I \preceq 0, \\ P \succeq I, \\ L - U_0 Q P^{-1} Q^\top U_0^\top \succeq 0, \\ X_0 Q = P, \\ \text{tr}(P) + \text{tr}(L) \leq \gamma, \end{cases} \end{aligned} \quad (16)$$

where U_0 and X_0, X_1 are the input and state data matrices defined according to (9). This relationship is formalized in the next lemma; the proof is along the lines of [8, Lemmas 2 and 3].

- Lemma 2.** 1) Let (γ, Q, P, L) be any feasible solution to (16), and define $K := U_0 Q P^{-1}$. Then $A + BK$ is Schur stable (i.e., all eigenvalues lie in the open unit disk).
2) Suppose that the rank condition (11) holds. For any K such that $A + BK$ is Schur stable, there exists a feasible solution (γ, Q, P, L) to (16) satisfying $K = U_0 Q P^{-1}$.

When only quantized state measurements are available, we replace X_0, X_1 with the quantized state data matrices $\hat{X}_0, \hat{X}_1$ defined in (9) and consider the following *quantized* SDP:

$$\begin{aligned} & \min_{(\gamma, Q, P, L, W)} \gamma \\ & \text{subject to} \\ & \begin{cases} \hat{X}_1 Q P^{-1} Q^\top \hat{X}_1^\top - P + I \preceq 0, \\ P \succeq I, \\ L - U_0 Q P^{-1} Q^\top U_0^\top \succeq 0, \\ \hat{X}_0 Q = P, \\ W - Q P^{-1} Q^\top \succeq 0, \\ \text{tr}(P) + \text{tr}(L) + \omega \text{tr}(W) \leq \gamma. \end{cases} \end{aligned} \quad (17)$$

As discussed in [8], the additional variable W and parameter $\omega > 0$ introduce a regularization term that enhances robustness to quantization errors. Larger values of ω favor solutions with smaller $\text{tr}(Q P^{-1} Q^\top)$, which typically facilitates satisfaction of the sufficient condition (19) in Theorem 1 below.

The data matrices satisfy

$$X_1 = A X_0 + B U_0, \quad \hat{X}_0 = X_0 + D_0, \quad \hat{X}_1 = X_1 + D_1,$$

which implies

$$\hat{X}_1 = A X_0 + B U_0 + D_1 = A \hat{X}_0 + B U_0 + (D_1 - A D_0).$$

Let

$$\Delta_0 := D_1 - A D_0 \in \mathbb{R}^{n \times T},$$

which captures the discrepancy between the true system dynamics and those implied by the quantized states. By (3) and (14), the columns of Δ_0 satisfy

$$\begin{aligned} \|d(k+1) - A d(k)\|_2 &\leq \|d(k+1)\|_2 + \|A\|_2 \|d(k)\|_2 \\ &\leq \sqrt{n} \|d(k+1)\|_\infty + \|A\|_2 \sqrt{n} \|d(k)\|_\infty \leq \sqrt{n}(\alpha + 1) \delta \end{aligned}$$

for all $k \in \{0, 1, \dots, T-1\}$. Using $\|\Delta_0\|_2 \leq \|\Delta_0\|_F$, we obtain

$$\|\Delta_0\|_2 \leq \sqrt{nT}(\alpha + 1) \delta =: \hat{\delta}. \quad (18)$$

Note that the quantization error bound δ , and hence $\hat{\delta}$, can be made arbitrarily small by increasing N_k , the number of quantization cells per dimension.

The following theorem shows that, any feasible solution to the quantized SDP (17) yields a stabilizing controller, provided that the solution satisfies a condition involving the quantization error bound and quantized state measurements.

Theorem 1. Let $(\bar{\gamma}, \bar{Q}, \bar{P}, \bar{L}, \bar{W})$ be any feasible solution to (17), and define $\bar{K} := U_0 \bar{Q} \bar{P}^{-1}$. If

$$\hat{\delta}^2 \|\bar{Q} \bar{P}^{-1} \bar{Q}^\top\|_2 + 2\hat{\delta} \|\hat{X}_1 \bar{Q} \bar{P}^{-1} \bar{Q}^\top\|_2 < 1, \quad (19)$$

where $\hat{\delta}$ is defined in (18), then $A + B \bar{K}$ is Schur stable. Moreover,

$$(A + B \bar{K}) \bar{P} (A + B \bar{K})^\top - \bar{P} \preceq -\eta_1 I, \quad (20)$$

where

$$\eta_1 := 1 - \hat{\delta}^2 \|\bar{Q} \bar{P}^{-1} \bar{Q}^\top\|_2 - 2\hat{\delta} \|\hat{X}_1 \bar{Q} \bar{P}^{-1} \bar{Q}^\top\|_2.$$

Proof. The proof is based on a data-driven representation of the closed-loop system. As $\bar{K} = U_0 \bar{Q} \bar{P}^{-1}$ and $\hat{X}_0 \bar{Q} \bar{P}^{-1} = I$, we have

$$\begin{aligned} A + B \bar{K} &= \begin{bmatrix} B & A \end{bmatrix} \begin{bmatrix} \bar{K} \\ I \end{bmatrix} = \begin{bmatrix} B & A \end{bmatrix} \begin{bmatrix} U_0 \\ \hat{X}_0 \end{bmatrix} \bar{Q} \bar{P}^{-1} \\ &= (A \hat{X}_0 + B U_0) \bar{Q} \bar{P}^{-1} = (\hat{X}_1 - \Delta_0) \bar{Q} \bar{P}^{-1}. \end{aligned}$$

Hence

$$\begin{aligned}
& (A + B\bar{K})\bar{P}(A + B\bar{K})^\top - \bar{P} \\
&= (\hat{X}_1 - \Delta_0)\bar{Q}\bar{P}^{-1}\bar{Q}^\top(\hat{X}_1 - \Delta_0)^\top - \bar{P} \\
&= \hat{X}_1\bar{Q}\bar{P}^{-1}\bar{Q}^\top\hat{X}_1^\top - \bar{P} + \bar{\Psi},
\end{aligned}$$

where

$$\bar{\Psi} := \Delta_0\bar{M}\Delta_0^\top - \hat{X}_1\bar{M}\Delta_0^\top - \Delta_0\bar{M}\hat{X}_1^\top, \quad \bar{M} := \bar{Q}\bar{P}^{-1}\bar{Q}^\top.$$

By the first constraint in (17),

$$(A + B\bar{K})\bar{P}(A + B\bar{K})^\top - \bar{P} \preceq \bar{\Psi} - I.$$

Using (18) and (19), we obtain

$$\begin{aligned}
\|\bar{\Psi}\|_2 &\leq \|\Delta_0\|_2^2\|\bar{M}\|_2 + 2\|\Delta_0\|_2\|\hat{X}_1\bar{M}\|_2 \\
&\leq \hat{\delta}^2\|\bar{M}\|_2 + 2\hat{\delta}\|\hat{X}_1\bar{M}\|_2 = 1 - \eta_1 < 1.
\end{aligned}$$

Hence $\bar{\Psi} \preceq (1 - \eta_1)I$, which implies (20). Therefore, $A + B\bar{K}$ is Schur stable. $\square$

In practice, the feedback gain $\bar{K}$ can be computed by solving the SDP (17) using input and quantized state data. Theorem 1 then provides a sufficient condition ensuring that $\bar{K}$ stabilizes the system, which can be verified directly from the collected data.

Our second theorem strengthens Theorem 1 by providing conditions under which the quantized SDP (17) is feasible and any optimal solution yields a stabilizing controller. The proof is provided in Appendix .

Theorem 2. *Suppose that the system (2) is controllable, and the input sequence $\{u(k)\}_{k=0}^{T-1}$ is PE of order $n+1$. If the quantization error bound δ is sufficiently small, then the rank condition (10) holds, the SDP (17) is feasible, and any optimal solution to (17) satisfies (19).*

A sufficient condition on δ is derived in the proof of Theorem 2. It depends on the unknown matrices A and B , and therefore cannot be computed offline or verified from the collected data. Nevertheless, the result provides an existence guarantee: for sufficiently fine quantization (i.e., sufficiently small δ), a stabilizing controller can be obtained. If the system can be reset to the same initial condition, the controller design stage can be repeated with progressively smaller values of δ , eventually yielding a stabilizing controller.

Theorem 2 is restricted to optimal solutions of the quantized SDP (17), in contrast to Theorem 1, which applies to all feasible solutions. This restriction arises because Theorem 2 is established by comparing the optimal value of (17) with that of its unquantized counterpart (16).

IV. QUANTIZED FEEDBACK STABILIZATION AND CONVERGENCE

Having obtained a stabilizing feedback gain from quantized data, we now address closed-loop stabilization using quantized state feedback.

A. Quantized state feedback implementation

At time T , suppose that a feasible solution $(\bar{\gamma}, \bar{Q}, \bar{P}, \bar{L}, \bar{W})$ to the quantized SDP (17) is obtained and (19) holds. Then by Theorem 1, the feedback gain $\bar{K} := U_0\bar{Q}\bar{P}^{-1}$ satisfies (20) and $A + B\bar{K}$ is Schur stable.

During the feedback stabilization stage, the first control input $u(T) = \bar{K}\hat{x}(T)$ is computed using a new quantized state measurement, obtained with the same quantization scheme but with updated parameters $\bar{E}_T$ and N_s replacing E_T and N_T , respectively. The quantizer update rules are then modified to use exponentially decreasing bounds and a fixed number of quantization cells:

$$E_{T+1} = \mu\bar{E}_T, \quad E_{k+1} = \mu E_k, \quad k = T+1, T+2, \dots \quad (21)$$

with a constant $\mu \in (0, 1)$ and

$$N_k = N_s, \quad k = T+1, T+2, \dots \quad (22)$$

The values of $\bar{E}_T, \mu$ and the condition on N_s are derived in Section IV-B. They are designed such that the state of the closed-loop system under quantized state feedback

$$x(k+1) = Ax(k) + B\bar{K}\hat{x}(k) \quad (23)$$

lies in the quantization range R_k defined in (5) for all $k > T$. Since the bounds E_k defined in (21) decay exponentially, the state converges exponentially to the origin.

Since the number of quantization cells is constant during this stage, the quantizer takes the form of a “zooming” quantizer [13]:

$$\hat{x}(k) = \mu_k q\left(\frac{x(k)}{\mu_k}\right),$$

where $q(\cdot)$ is the quantizer with bound $\bar{E}_T$ and N_s quantization cells per dimension, and

$$\mu_k := \mu^{k-T}, \quad k = T, T+1, T+2, \dots$$

is the zoom variable.

B. Exponential convergence

We establish exponential convergence by constructing nested sets that contract under the closed-loop dynamics, based on sublevel sets of

$$V(x) = x^\top \bar{P}x, \quad (24)$$

where $\bar{P}$ is from the feasible solution to (17) obtained at time T .

We first construct a sublevel set of V containing $x(T)$ and enlarge the quantization range to a hypercube containing this set. Define

$$\begin{aligned} R(E) &:= \{x \in \mathbb{R}^n : \|x\|_\infty \leq E\}, \\ S(E) &:= \{x \in \mathbb{R}^n : V(x) \leq \underline{\lambda}(\bar{P})E^2\}. \end{aligned}$$

For all $x \in \mathbb{R}^n$, using

$$\underline{\lambda}(\bar{P})\|x\|_2^2 \leq x^\top \bar{P}x \leq \bar{\lambda}(\bar{P})\|x\|_2^2 \quad (25)$$

together with (1), we obtain

$$\underline{\lambda}(\bar{P})\|x\|_\infty^2 \leq V(x) \leq n\bar{\lambda}(\bar{P})\|x\|_\infty^2.$$

Then $S(E) \subset R(E)$ for all $E > 0$. Moreover, $x(T) \in R_T = R(E_T)$ implies

$$x(T) \in S(\bar{E}_T) \subset R(\bar{E}_T),$$

where

$$\bar{E}_T := \sqrt{\frac{n\bar{\lambda}(\bar{P})}{\underline{\lambda}(\bar{P})}} E_T, \quad (26)$$

and $R(\bar{E}_T)$ is the new quantization range at T .

Theorem 3. *If the number of quantization cells satisfies*

$$N_s^2 > \frac{n(\bar{\lambda}(\bar{P})(\theta + \sqrt{(\phi + \varepsilon)/\eta_1})^2 - \varepsilon)}{\underline{\lambda}(\bar{P})} \quad (27)$$

with $\varepsilon > 0$ and

$$\begin{aligned} \theta &:= \frac{\beta(\alpha + \beta\|\bar{K}\|_2)\|\bar{P}\|_2\|\bar{K}\|_2}{\eta_1}, \\ \phi &:= \beta^2\|\bar{P}\|_2\|\bar{K}\|_2^2 + \eta_1\theta^2, \end{aligned}$$

where $\alpha, \beta > 0$ are as in (3) and $\eta_1 \in (0, 1)$ is as in (20), then

$$\|x(k)\|_\infty \leq \mu^{k-T} \bar{E}_T \quad \forall k \geq T, \quad (28)$$

where

$$\mu := \sqrt{\max\left\{\frac{n\bar{\lambda}(\bar{P})(\theta + \sqrt{(\phi + \varepsilon)/\eta_1})^2}{\underline{\lambda}(\bar{P})N_s^2}, 1\right\}} - \frac{n\varepsilon}{\underline{\lambda}(\bar{P})N_s^2} \quad (29)$$

satisfies $\mu \in (0, 1)$.

The proof of Theorem 3 relies on the following lemma.

Lemma 3. Suppose that at time $k \geq T$,

$$x(k) \in S(E_k) \subset R(E_k),$$

where $R(E_k)$ is the quantization range. Then

$$x(k+1) \in S(\mu E_k) \subset R(\mu E_k), \quad (30)$$

where μ is defined in (29).

Proof. For brevity, we let $x = x(k)$, $x^+ = x(k+1)$, $V = V(x)$, $V^+ = V(x^+)$, $d = d(k)$, and $E = E_k$ in the proof of Lemma 3. By (23),

$$\begin{aligned} V^+ &= ((A + B\bar{K})x + B\bar{K}d)^\top \bar{P}((A + B\bar{K})x + B\bar{K}d) \\ &= x^\top (A + B\bar{K})^\top \bar{P}(A + B\bar{K})x \\ &\quad + 2x^\top (A + B\bar{K})^\top \bar{P}B\bar{K}d + d^\top \bar{K}^\top B^\top \bar{P}B\bar{K}d. \end{aligned}$$

By (20),

$$x^\top (A + B\bar{K})^\top \bar{P}(A + B\bar{K})x \leq V - \eta_1 \|x\|_2^2.$$

Applying (3) and (6), we obtain

$$\begin{aligned} x^\top (A + B\bar{K})^\top \bar{P}B\bar{K}d &\leq \|(A + B\bar{K})^\top \bar{P}B\bar{K}\|_2 \|x\|_2 \|d\|_2 \\ &\leq \beta(\alpha + \beta\|\bar{K}\|_2)\|\bar{P}\|_2\|\bar{K}\|_2\|x\|_2\sqrt{n}\|d\|_\infty \\ &\leq \eta_1\theta\|x\|_2\frac{\sqrt{n}E}{N_s}, \end{aligned}$$

and

$$\begin{aligned} d^\top \bar{K}^\top B^\top \bar{P}B\bar{K}d &\leq \|\bar{K}^\top B^\top \bar{P}B\bar{K}\|_2 \|d\|_2^2 \\ &\leq \beta^2\|\bar{P}\|_2\|\bar{K}\|_2^2 n\|d\|_\infty^2 \\ &\leq \beta^2\|\bar{P}\|_2\|\bar{K}\|_2^2 \frac{nE^2}{N_s^2}. \end{aligned}$$

Hence

$$\begin{aligned} V^+ &\leq V - \eta_1\|x\|_2^2 + 2\eta_1\theta\|x\|_2\frac{\sqrt{n}E}{N_s} \\ &\quad + \beta^2\|\bar{P}\|_2\|\bar{K}\|_2^2 \frac{nE^2}{N_s^2} \\ &= V - \eta_1\left(\|x\|_2 - \frac{\theta\sqrt{n}E}{N_s}\right)^2 + \frac{\phi nE^2}{N_s^2}. \end{aligned}$$

If

$$\|x\|_2 \geq \frac{(\theta + \sqrt{(\phi + \varepsilon)/\eta_1})\sqrt{n}E}{N_s},$$

then since $x \in S(E)$,

$$V^+ \leq V - \frac{\varepsilon nE^2}{N_s^2} \leq \left(\underline{\lambda}(\bar{P}) - \frac{n\varepsilon}{N_s^2}\right)E^2 \leq \underline{\lambda}(\bar{P})(\mu E)^2.$$

Thus $x^+ \in S(\mu E) \subset R(\mu E)$.

Otherwise,

$$\|x\|_2 < \frac{(\theta + \sqrt{(\phi + \varepsilon)/\eta_1})\sqrt{n}E}{N_s},$$

so by (25),

$$\begin{aligned} V^+ &\leq (\bar{\lambda}(\bar{P}) - \eta_1)\|x\|_2^2 + 2\eta_1\theta\|x\|_2\frac{\sqrt{n}E}{N_s} \\ &\quad + \beta^2\|\bar{P}\|_2\|\bar{K}\|_2^2\frac{nE^2}{N_s^2}, \end{aligned}$$

which is nondecreasing in $\|x\|_2$ since $\bar{P} - \eta_1 I \succeq 0$. Therefore,

$$\begin{aligned} V^+ &\leq \bar{\lambda}(\bar{P})\|x\|_2^2 - \eta_1\left(\|x\|_2 - \frac{\theta\sqrt{n}E}{N_s}\right)^2 + \frac{\phi nE^2}{N_s^2} \\ &\leq \frac{\bar{\lambda}(\bar{P})(\theta + \sqrt{(\phi + \varepsilon)/\eta_1})^2 nE^2}{N_s^2} - \frac{\varepsilon nE^2}{N_s^2} \\ &= \frac{n(\bar{\lambda}(\bar{P})(\theta + \sqrt{(\phi + \varepsilon)/\eta_1})^2 - \varepsilon)}{N_s^2} E^2 \leq \underline{\lambda}(\bar{P})(\mu E)^2. \end{aligned}$$

Thus $x^+ \in S(\mu E) \subset R(\mu E)$ still holds. $\square$

Proof of Theorem 3. For all $k \geq T$, iterating Lemma 3 yields

$$x(k) \in R(\mu^{k-T} \bar{E}_T),$$

which implies (28). Under (27), we have $\mu \in (0, 1)$. $\square$

V. SIMULATION RESULTS

We illustrate the proposed approach on the linear system (2) with

$$A = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

which is unstable in open-loop. The dimensions are $n = 2$ and $m = 1$. Following (8), the controller design stage lasts $T = m(n + 1) + n = 5$ steps. The feedback stabilization stage is simulated for 30 steps.

During the controller design stage, the input sequence is PE of order $n + 1 = 3$ and satisfies $\|u(k)\|_2 \leq \bar{u} = 0.5$. The initial state is $x_0 = [0.21 \ 0.32]^\top$. We apply a quantizer with $E_0 = 2$, $\delta = 0.1$, $\alpha = 3 > \|A\|_2$, and $\beta = 1.5 > \|B\|_2$. By (13) and the data-dependent update rule (15), the number of quantization cells per dimension is

$$N_k = 10, 10, 18, 9, 16, 25.$$

The collected data satisfy the rank condition (10). Consistent with Theorems 1 and 2, the SDP (17) is feasible. An optimal solution is computed, which satisfies the condition (19). The resulting feedback gain is

$$\bar{K} = U_0 \bar{Q} \bar{P}^{-1} = \begin{bmatrix} -0.8488 & -0.1726 \end{bmatrix},$$

and $A + B\bar{K}$ is Schur stable.

During the feedback stabilization stage, the quantization range is updated according to (26), yielding $\bar{E}_T = 11.84$. From the theoretical condition (27), the number of quantization cells is required to satisfy $N_s > 4627$. In simulation, exponential convergence is achieved with significantly fewer quantization cells. Specifically, using $N_s = 20$, we observe convergence with rate $\mu = 0.95$. This gap highlights the conservativeness of the worst-case analysis.

Fig. 1 illustrates the evolution of the state norm $\|x(k)\|_\infty$ and the quantization bound E_k . During the controller design stage (steps 0–5), the state is driven by the PE input, which may cause the quantization range to expand while maintaining bounded quantization error. During the feedback stabilization stage (steps 6–35), the

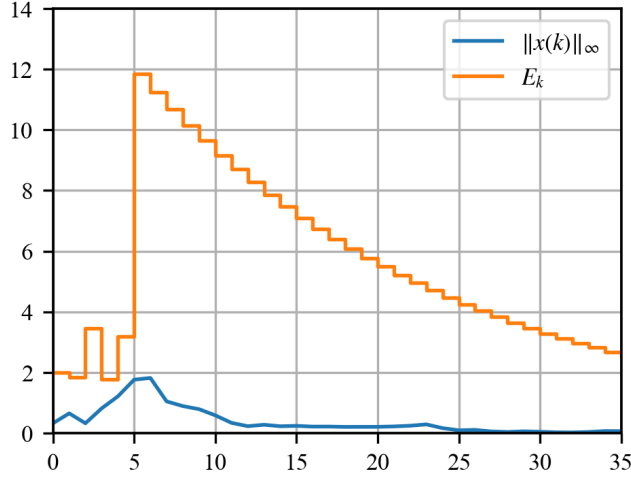


Fig. 1: State norm $\|x(k)\|_\infty$ (blue) and quantization bound E_k (orange) over 35 simulation steps. The transition from controller design stage to feedback stabilization stage occurs at step 5.

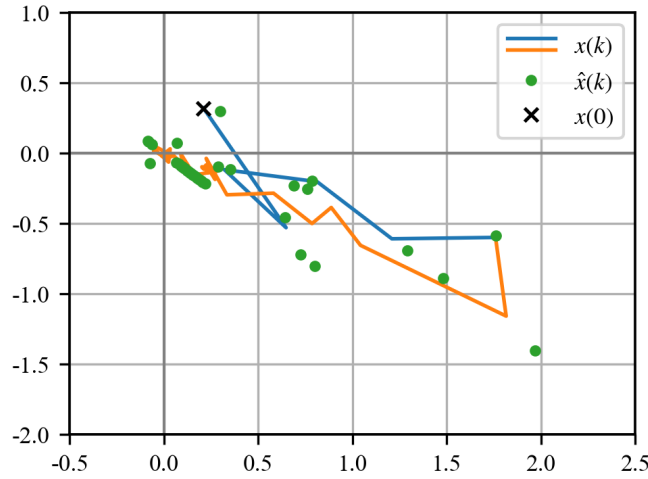


Fig. 2: State-space trajectory $x(k)$ (blue: controller design stage; orange: feedback stabilization stage) and quantized measurements $\hat{x}(k)$ (green). The initial state is indicated by a cross.

quantization range is first enlarged due to the Lyapunov-based design and then contracts exponentially, yielding an exponentially decaying state bound consistent with Theorem 3.

Fig. 2 shows the corresponding state-space trajectory $x(k)$ and quantized measurements $\hat{x}(k)$. The trajectory transitions from exploration during the controller design stage to contraction during the feedback stabilization stage, illustrating the effectiveness of the proposed quantized control method.

Overall, the simulation results demonstrate that the proposed approach successfully stabilizes the unknown system despite quantization errors in both the controller design and feedback stabilization stages. Moreover, the required number of quantization cells in practice is significantly smaller than the theoretical bounds, indicating conservativeness of the analysis.

VI. CONCLUSION

This paper proposed a data-driven framework for stabilizing unknown discrete-time linear systems using only quantized state measurements. A two-stage approach was developed, where a stabilizing feedback gain is computed from quantized data via an SDP, and exponential convergence is achieved under a Lyapunov-based quantization scheme. The number of quantization levels is finite during controller design and fixed during stabilization.

Future work includes extending the proposed framework to continuous-time LTI systems and switched systems, as well as improving the quantization scheme to reduce the required number of quantization cells while preserving

stability guarantees.

APPENDIX PROOF OF THEOREM 2

The proof consists of three main steps.

A. Establishing the rank condition (10)

Let

$$\hat{Z} = \begin{bmatrix} U_0 \\ \hat{X}_0 \end{bmatrix}, \quad Z = \begin{bmatrix} U_0 \\ X_0 \end{bmatrix}.$$

In this step, we derive a sufficient condition under which (10) holds, that is, $\hat{Z}$ has full row rank, equivalently, its smallest singular value $\underline{\sigma}(\hat{Z}) > 0$.

Since

$$\hat{Z} = Z + \begin{bmatrix} 0 \\ D_0 \end{bmatrix},$$

Weyl's inequality for singular values implies

$$\underline{\sigma}(\hat{Z}) \geq \underline{\sigma}(Z) - \|D_0\|_2.$$

We first bound $\underline{\sigma}(Z)$. Since the input sequence $\{u(k)\}_{k=0}^{T-1}$ is PE of order $n+1$, the Hankel matrix $\mathcal{U}_{0,n+1,T-n}$ defined in Definition 1 has full row rank $m(n+1)$, or equivalently, $\underline{\sigma}(\mathcal{U}_{0,n+1,T-n}) > 0$. Moreover, since the pair (A, B) is controllable, a robust extension of Lemma 1 in [27, Lemma 3.2] yields

$$\underline{\sigma}(Z) \geq \frac{\rho_0}{\sqrt{n+1}} \underline{\sigma}(\mathcal{U}_{0,n+1,T-n}),$$

where $\rho_0 > 0$ is a constant determined by the system matrices A and B . Next, we bound $\|D_0\|_2$. Using $\|D_0\|_2 \leq \|D_0\|_F$ and the quantization error bound (14), we obtain

$$\|D_0\|_2 \leq \sqrt{\sum_{k=0}^{T-1} \|d(k)\|_2^2} \leq \sqrt{\sum_{k=0}^{T-1} n \|d(k)\|_\infty^2} \leq \sqrt{nT} \delta.$$

Combining these bounds yields

$$\underline{\sigma}(\hat{Z}) \geq \frac{\rho_0}{\sqrt{n+1}} \underline{\sigma}(\mathcal{U}_{0,n+1,T-n}) - \sqrt{nT} \delta.$$

Therefore, if

$$\delta < \frac{\rho_0}{\sqrt{(n+1)nT}} \underline{\sigma}(\mathcal{U}_{0,n+1,T-n}) =: \delta_0, \tag{31}$$

then

$$\underline{\sigma}(\hat{Z}) \geq \sqrt{nT}(\delta_0 - \delta) > 0, \tag{32}$$

which establishes (10).

B. Establishing feasibility of the quantized SDP (17)

In this step, we derive a sufficient condition under which a feasible solution to (17) can be constructed.

Suppose that (31) holds. Consider the following SDP:

$$\begin{aligned} & \min_{(\gamma, K, P, L)} \gamma \\ & \text{subject to} \\ & \begin{cases} (A + BK)P(A + BK)^\top - P + I \preceq 0, \\ P \succeq I, \\ L - KPK^\top \succeq 0, \\ \text{tr}(P) + \text{tr}(L) \leq \gamma. \end{cases} \end{aligned} \quad (33)$$

Since (A, B) is controllable, there exists a feedback gain K_* such that $A + BK_*$ is Schur stable. Let $P_* \succ 0$ be the solution to the discrete-time Lyapunov equation

$$(A + BK_*)P_*(A + BK_*)^\top - P_* + I = 0,$$

and define

$$L_* := K_*P_*K_*^\top, \quad \gamma_* := \text{tr}(P_*) + \text{tr}(L_*), \quad Q_* := G_{K_*}P_*,$$

where

$$G_{K_*} := \hat{Z}^+ \begin{bmatrix} K_* \\ I \end{bmatrix},$$

and $\hat{Z}^+$ is the *pseudoinverse* of $\hat{Z}$ and satisfies $\hat{Z}^+ = \hat{Z}^\top (\hat{Z}\hat{Z}^\top)^{-1}$ under the rank condition (10). Then $(\gamma_*, K_*, P_*, L_*)$ is a feasible solution to (33).

We now construct a feasible solution to (17). Let

$$W_* := Q_*P_*^{-1}Q_*^\top = \hat{Z}^+\Xi(\hat{Z}^+)^\top,$$

where

$$\Xi := \begin{bmatrix} K_* \\ I \end{bmatrix} P_* \begin{bmatrix} K_* \\ I \end{bmatrix}^\top$$

depends on the system matrices A and B . Consider the tuple $(\gamma_* + \omega \text{tr}(W_*), Q_*, P_*, L_*, W_*)/\eta_2$, where $\eta_2 \in (0, 1)$ is a constant to be specified later. By construction, this tuple satisfies all constraints in (17) except possibly the first one, which we verify next. Using

$$\begin{aligned} (\hat{X}_1 - \Delta_0)Q_* &= (A\hat{X}_0 + BU_0)Q_* \\ &= \begin{bmatrix} B & A \end{bmatrix} \hat{Z}Q_* = (A + BK_*)P_*, \end{aligned}$$

we obtain

$$\begin{aligned} & \hat{X}_1Q_*P_*^{-1}Q_*^\top\hat{X}_1^\top \\ &= (\hat{X}_1 - \Delta_0)Q_*P_*^{-1}Q_*^\top(\hat{X}_1 - \Delta_0)^\top - \Psi_* \\ &= (A + BK_*)P_*(A + BK_*)^\top - \Psi_* \\ &= P_* - I - \Psi_*, \end{aligned}$$

where

$$\Psi_* := \Delta_0W_*\Delta_0^\top - \hat{X}_1W_*\Delta_0^\top - \Delta_0W_*\hat{X}_1^\top.$$

Hence

$$\frac{1}{\eta_2}\hat{X}_1Q_*P_*^{-1}Q_*^\top\hat{X}_1^\top - \frac{1}{\eta_2}P_* + I = \frac{(\eta_2 - 1)I - \Psi_*}{\eta_2}.$$

We rewrite $-\Psi_*$ as

$$\begin{aligned} -\Psi_* &= -\Delta_0 W_* \Delta_0^\top + \hat{X}_1 W_* \Delta_0^\top + \Delta_0 W_* \hat{X}_1^\top \\ &= \Delta_0 W_* \Delta_0^\top + (\hat{X}_1 - \Delta_0) W_* \Delta_0^\top + \Delta_0 W_* (\hat{X}_1 - \Delta_0)^\top, \end{aligned}$$

where $W_* := Q_* P_*^{-1} Q_*^\top = \hat{Z}^+ \Xi (\hat{Z}^+)^\top$ and

$$\begin{aligned} (\hat{X}_1 - \Delta_0) W_* &= (\hat{X}_1 - \Delta_0) Q_* P_*^{-1} Q_*^\top \\ &= (A + B K_*) Q_*^\top = \begin{bmatrix} B & A \end{bmatrix} \Xi (\hat{Z}^+)^\top. \end{aligned}$$

Moreover, the derivation of (18) also yields

$$\|\Delta_0\|_2 \leq \sqrt{nT}(\|A\|_2 + 1)\delta, \quad (34)$$

and (32) implies

$$\|\hat{Z}^+\|_2 = \frac{1}{\underline{\sigma}(\hat{Z})} \leq \frac{1}{\sqrt{nT}(\delta_0 - \delta)}. \quad (35)$$

Therefore,

$$\begin{aligned} \|-\Psi_*\|_2 &\leq \|\Xi\|_2 \|\hat{Z}^+\|_2^2 \|\Delta_0\|_2^2 \\ &\quad + 2\|\Xi\|_2 \|\hat{Z}^+\|_2 \|\Delta_0\|_2 \sqrt{\|A\|_2^2 + \|B\|_2^2} \\ &\leq \|\Xi\|_2 \left(\frac{(\|A\|_2 + 1)\delta}{\delta_0 - \delta} \right)^2 \\ &\quad + 2\|\Xi\|_2 \frac{(\|A\|_2 + 1)\delta}{\delta_0 - \delta} \sqrt{\|A\|_2^2 + \|B\|_2^2} \\ &= \|\Xi\|_2 \left(\frac{(\|A\|_2 + 1)\delta}{\delta_0 - \delta} + \sqrt{\|A\|_2^2 + \|B\|_2^2} \right)^2 \\ &\quad - \|\Xi\|_2 (\|A\|_2^2 + \|B\|_2^2). \end{aligned}$$

Bounding the above expression by $1 - \eta_2$ and solving for δ , we obtain the following sufficient condition. For any $\eta_2 \in (0, 1)$, if

$$\delta \leq \frac{\delta_0}{1 + \frac{\|A\|_2 + 1}{\sqrt{\|A\|_2^2 + \|B\|_2^2 + \frac{1-\eta_2}{\|\Xi\|_2} - \sqrt{\|A\|_2^2 + \|B\|_2^2}}}} =: \delta_1 \quad (36)$$

(which implies that (31) holds), then $(\eta_2 - 1)I - \Psi_* \preceq 0$ and thus

$$\frac{1}{\eta_2} \hat{X}_1 Q_* P_*^{-1} Q_*^\top \hat{X}_1^\top - \frac{1}{\eta_2} P_* + I \preceq 0.$$

Consequently, $(\gamma_* + \omega \text{tr}(W_*), Q_*, P_*, L_*, W_*)/\eta_2$ satisfies the first constraint in (17) and is thus a feasible solution to (17).

C. Establishing (19) for any optimal solution to (17)

Let $(\bar{\gamma}, \bar{Q}, \bar{P}, \bar{L}, \bar{W})$ be an arbitrary optimal solution to (17). In this step, we derive a sufficient condition, independent of the particular optimal solution, under which (19) holds.

Suppose that (36) holds and a feasible solution $(\gamma_* + \omega \text{tr}(W_*), Q_*, P_*, L_*, W_*)/\eta_2$ to (17) is constructed as in Step B. Let $\bar{K} := U_0 \bar{Q} \bar{P}^{-1}$. Using arguments similar to those in the proof of Theorem 1, we obtain

$$A + B \bar{K} = (\hat{X}_1 - \Delta_0) \bar{Q} \bar{P}^{-1}.$$

Substituting this and (18) into (19) yields

$$\begin{aligned}
& \hat{\delta}^2 \|\bar{Q}\bar{P}^{-1}\bar{Q}^\top\|_2 + 2\hat{\delta} \|\hat{X}_1 \bar{Q}\bar{P}^{-1}\bar{Q}^\top\|_2 \\
& \leq 3\hat{\delta}^2 \|\bar{Q}\bar{P}^{-1}\bar{Q}^\top\|_2 + 2\hat{\delta} \|(\hat{X}_1 - \Delta_0) \bar{Q}\bar{P}^{-1}\bar{Q}^\top\|_2 \\
& \leq 3\hat{\delta}^2 \|\bar{Q}\bar{P}^{-1}\bar{Q}^\top\|_2 + 2\hat{\delta} \|(A + B\bar{K}) \bar{Q}^\top\|_2 \\
& = 3\hat{\delta}^2 \|\bar{Q}\bar{P}^{-1}\bar{Q}^\top\|_2 + 2\hat{\delta} \left\| \begin{bmatrix} B & A \end{bmatrix} \begin{bmatrix} K \\ I \end{bmatrix} \bar{Q}^\top \right\|_2 \\
& \leq 3\hat{\delta}^2 \|\bar{Q}\bar{P}^{-1}\bar{Q}^\top\|_2 + 2\hat{\delta} \|\bar{Q}\|_2 \sqrt{\|\bar{K}\|_2^2 + 1} \sqrt{\|A\|_2^2 + \|B\|_2^2}.
\end{aligned}$$

Since $(\bar{\gamma}, \bar{Q}, \bar{P}, \bar{L}, \bar{W})$ is optimal for (17), we have

$$\bar{\gamma} = \text{tr}(\bar{P}) + \text{tr}(\bar{K}\bar{P}\bar{K}^\top) + \omega \text{tr}(\bar{Q}\bar{P}^{-1}\bar{Q}^\top).$$

Moreover, since $(\gamma_* + \omega \text{tr}(W_*), Q_*, P_*, L_*, W_*)/\eta_2$ is feasible for (17), optimality yields

$$\text{tr}(\bar{P}) + \text{tr}(\bar{K}\bar{P}\bar{K}^\top) + \omega \text{tr}(\bar{Q}\bar{P}^{-1}\bar{Q}^\top) \leq \frac{\gamma_* + \omega \text{tr}(W_*)}{\eta_2}.$$

We first bound $\text{tr}(W_*)$. Using $\Xi \succeq 0$ and (35), we obtain

$$\begin{aligned}
\text{tr}(W_*) &= \text{tr}(\hat{Z}^+ \Xi (\hat{Z}^+)^{\top}) \\
&\leq \text{tr}(\Xi) \bar{\lambda}((\hat{Z}^+)^{\top} \hat{Z}^+) \\
&= \text{tr}(\Xi) \|\hat{Z}^+\|_2^2 \\
&\leq \frac{\text{tr}(\Xi)}{nT(\delta_0 - \delta)^2} \leq \frac{\text{tr}(\Xi)}{nT(\delta_0 - \delta_1)^2},
\end{aligned}$$

and thus

$$\gamma_* + \omega \text{tr}(W_*) \leq \gamma_* + \frac{\omega \text{tr}(\Xi)}{nT(\delta_0 - \delta_1)^2} =: \chi.$$

Next, using $\bar{P} \succeq I$, we obtain

$$\begin{aligned}
\text{tr}(\bar{P}) + \text{tr}(\bar{K}\bar{P}\bar{K}^\top) &= \text{tr} \left(\begin{bmatrix} \bar{K} \\ I \end{bmatrix} \bar{P} \begin{bmatrix} \bar{K} \\ I \end{bmatrix}^\top \right) \\
&\geq \underline{\lambda}(\bar{P}) \text{tr} \left(\begin{bmatrix} \bar{K} \\ I \end{bmatrix}^\top \begin{bmatrix} \bar{K} \\ I \end{bmatrix} \right) \\
&= \underline{\lambda}(\bar{P}) \text{tr}(\bar{K}^\top \bar{K} + I) \\
&\geq \underline{\lambda}(\bar{P})(\|\bar{K}\|_2^2 + 1) \geq \|\bar{K}\|_2^2 + 1.
\end{aligned}$$

and

$$\begin{aligned}
\text{tr}(\bar{Q}\bar{P}^{-1}\bar{Q}^\top) &\geq \underline{\lambda}(\bar{P}^{-1}) \text{tr}(\bar{Q}^\top \bar{Q}) \\
&\geq \underline{\lambda}(\bar{P}^{-1}) \|\bar{Q}\|_2^2 = \frac{\|\bar{Q}\|_2^2}{\|\bar{P}\|_2}.
\end{aligned}$$

Combining these bounds yields

$$\begin{aligned}
\|\bar{Q}\bar{P}^{-1}\bar{Q}^\top\|_2 &\leq \text{tr}(\bar{Q}\bar{P}^{-1}\bar{Q}^\top) \leq \frac{\chi}{\eta_2 \omega}, \\
\|\bar{K}\|_2^2 + 1 &\leq \text{tr}(\bar{P}) + \text{tr}(\bar{K}\bar{P}\bar{K}^\top) \leq \frac{\chi}{\eta_2},
\end{aligned}$$

and

$$\begin{aligned}\|\bar{Q}\|_2 &\leq \sqrt{\|\bar{P}\|_2 \operatorname{tr}(\bar{Q}\bar{P}^{-1}\bar{Q}^\top)} \\ &\leq \sqrt{\operatorname{tr}(\bar{P}) \operatorname{tr}(\bar{Q}\bar{P}^{-1}\bar{Q}^\top)} \\ &\leq \frac{\operatorname{tr}(\bar{P}) + \omega \operatorname{tr}(\bar{Q}\bar{P}^{-1}\bar{Q}^\top)}{2\sqrt{\omega}} \leq \frac{\chi}{2\eta_2\sqrt{\omega}}.\end{aligned}$$

Therefore,

$$\begin{aligned}&\delta^2 \|\bar{Q}\bar{P}^{-1}\bar{Q}^\top\|_2 + 2\hat{\delta} \|\hat{X}_1 \bar{Q}\bar{P}^{-1}\bar{Q}^\top\|_2 \\ &\leq 3\hat{\delta}^2 \|\bar{Q}\bar{P}^{-1}\bar{Q}^\top\|_2 + 2\hat{\delta} \|\bar{Q}\|_2 \sqrt{\|\bar{K}\|_2^2 + 1} \sqrt{\|A\|_2^2 + \|B\|_2^2} \\ &\leq \frac{3\chi\hat{\delta}^2}{\eta_2\omega} + \frac{\chi\hat{\delta}}{\eta_2} \sqrt{\frac{\chi}{\eta_2\omega}} \sqrt{\|A\|_2^2 + \|B\|_2^2} \\ &= \frac{3\chi}{\eta_2\omega} \left(\hat{\delta} + \frac{1}{6} \sqrt{\frac{\chi\omega}{\eta_2}} \sqrt{\|A\|_2^2 + \|B\|_2^2} \right)^2 - \frac{\chi^2}{12\eta_2^2} (\|A\|_2^2 + \|B\|_2^2).\end{aligned}$$

Bounding the above expression by 1 and solving for $\delta = \hat{\delta}/(\sqrt{nT}(\alpha + 1))$, we obtain the following sufficient condition. For any $\eta_2 \in (0, 1)$, if (36) holds and

$$\begin{aligned}\delta &< \frac{1}{6\sqrt{nT}(\alpha + 1)} \sqrt{\frac{\chi\omega}{\eta_2}} \left(\sqrt{\|A\|_2^2 + \|B\|_2^2} + \frac{12\eta_2^2}{\chi^2} \right. \\ &\quad \left. - \sqrt{\|A\|_2^2 + \|B\|_2^2} \right) =: \delta_2,\end{aligned}\tag{37}$$

then (19) holds.

Combining the three steps, Theorem 2 holds if

$$\delta < \min\{\delta_1, \delta_2\},$$

where δ_1 and δ_2 are defined in (36) and (37), respectively.

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